

Trainer Debrief — Lab 26

FMEA, Pilot Design and Implementation Planning — In Depth

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: De-risk and pilot your capstone improvement before full rollout (A5).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-26-fmea-worksheet.xlsx

Scored realistically, the two highest RPNs are 'Fixture clamp pressure drifts during shift' (S=8, O=7, D=8, RPN=448) and 'Electrode tip wears past service limit' (S=7, O=6, D=8, RPN=336) — both scoring high on DETECTION because nothing catches them before the burst test. That is the argument for the Lab 27 control plan: a clamp-pressure interlock and a tip stroke counter attack Detection, which is the cheapest of the three to improve. Any mode with S=9 or 10 is actioned regardless of RPN.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your FMEA is re-scored post-mitigation with the severity override applied, pilot success criteria were defined before starting, and the improvement is tested statistically against the documented baseline.

Where this sits in the capstone

Re-derive FMEA in full, then design the pilot that proves your improvement works before it is scaled. A Black Belt never rolls out an unpiloted change across sites — the cost of being wrong at scale is exactly what the pilot buys down. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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